

SAAKSHYA

साक्ष्य · evidence

Federated CCTV Intelligence and Evidence Fabric

Project report · Gujarat Police Innovation Challenge 2026 · Sentinel Camera Grid
DAU (DA-IICT) · 28 September 2026

1+2+3+4 all four models built	1,155,325 government observations	1,101 plate reads delivered	1,582 tests passing
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Figures are labelled MEASURED, MODELLED, DEMO or DESIGNED in the full submission; see docs/FINAL_SUBMISSION.md and reports/FINAL_SUBMISSION_CERTIFICATION.md in the repository.

1. Synopsis / Abstract

SAAKSHYA (साक्ष्य, "evidence") is a federated CCTV intelligence and evidence platform for Gujarat Police, built for the Sentinel Camera Grid challenge as a hybrid of all four reference models, 1 + 2 + 3 + 4, all built. Model 1 onboards any existing camera into one registry and GIS with measured capability grades. Model 2 views the government grid directly over RTSP and WebRTC (WHEP). Model 3 federates departmental VMS systems through adapters. Model 4 runs central AI analytics on the cameras that measurably deserve it; it ran live on the government grid on 28 September 2026 (four deep-inference slots, 4,465 observations, 200 plate reads).

Every vehicle sighting becomes a timestamped, geolocated, hash-chained observation. Officers search by plate, partial or near-match plate and attributes, follow a vehicle's movement on the map, receive automatic real-time watchlist alerts, and export sealed evidence. Every search requires a case number and a stated purpose and is written to a tamper-evident audit log. Video stays at the camera; only about 1.3 kB of metadata per observation moves, which keeps a statewide rollout to about 80,000 cameras affordable.

2. Literature Review

City-surveillance programmes (Safe City projects, command-and-control centres) have mostly centralised video from one vendor's VMS. That model does not fit an estate spread over 26 departments, many vendors and two decades of hardware: at 2 Mbps per camera, 80,000 cameras need about 160 Gbps and 30 days of storage need about 52 PB. Federated and edge-first designs keep video at the source and move only events and metadata.

Object detection has moved from two-stage detectors to real-time single-stage and transformer detectors: the YOLO family (for example YOLOv9, Wang et al., 2024) and RT-DETR (Zhao et al., "DETRs Beat YOLOs on Real-time Object Detection", CVPR 2024), with RT-DETRv2 improving it further. Multi-object tracking by detection (for example ByteTrack, Zhang et al., ECCV 2022) associates low- and high-confidence boxes to keep identities through occlusion.

For number plates, open ANPR stacks (for example fast-alpr) pair a plate detector with an OCR model. Indian plates are hard for generic OCR: mixed formats, two-line plates on two-wheelers, dirt and low resolution. OCR models such as PaddleOCR's PP-OCR family (Du et al., 2020) can be fine-tuned on Indian plates. Awiros-ANPR-OCR is one such model, trained on about 558,000 Indian plates. Voting across the frames of one track reduces single-frame misreads.

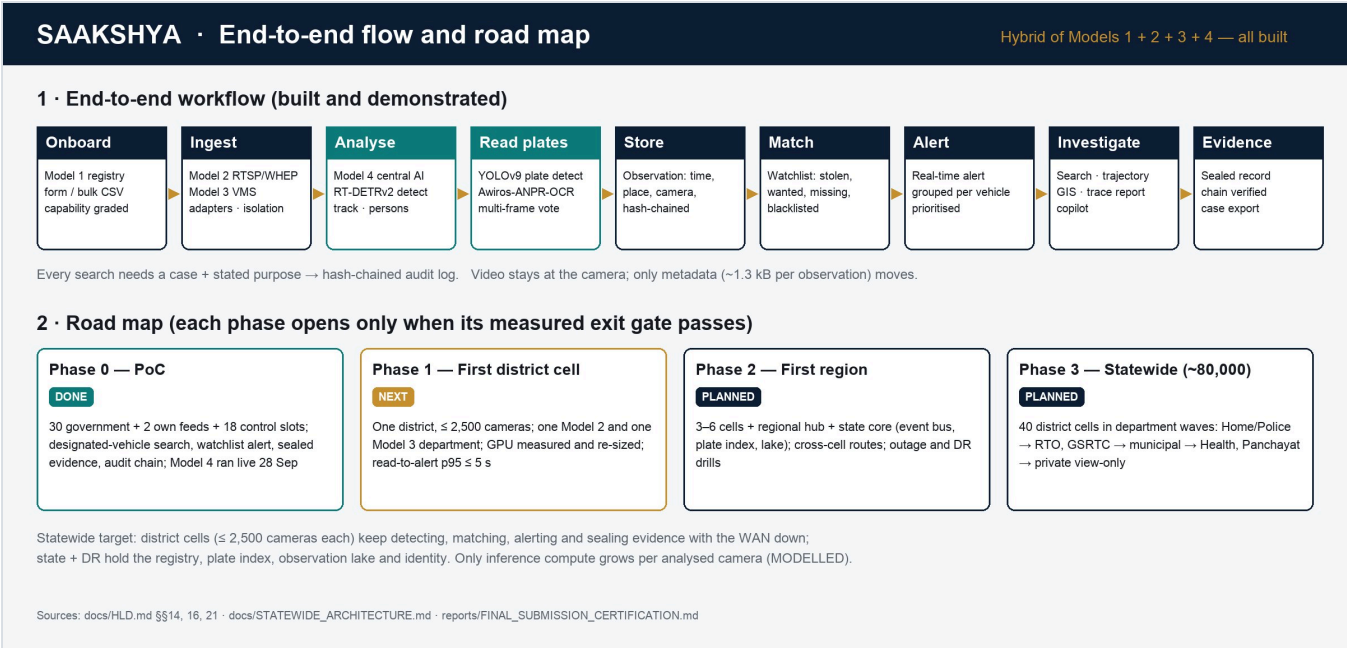
National systems (CCTNS/ICJS, VAHAN and SARATHI) hold the watchlist and vehicle records that alerts must be checked against. The Bharatiya Sakshya Adhiniyam 2023 (s.63) sets the certificate requirements for electronic evidence, and the Digital Personal Data Protection Act 2023 governs personal data. Our design follows these: sealed and hashed records, a certificate draft that humans sign, purpose-bound and audited access, and no face recognition on government data.

3. Your Approach

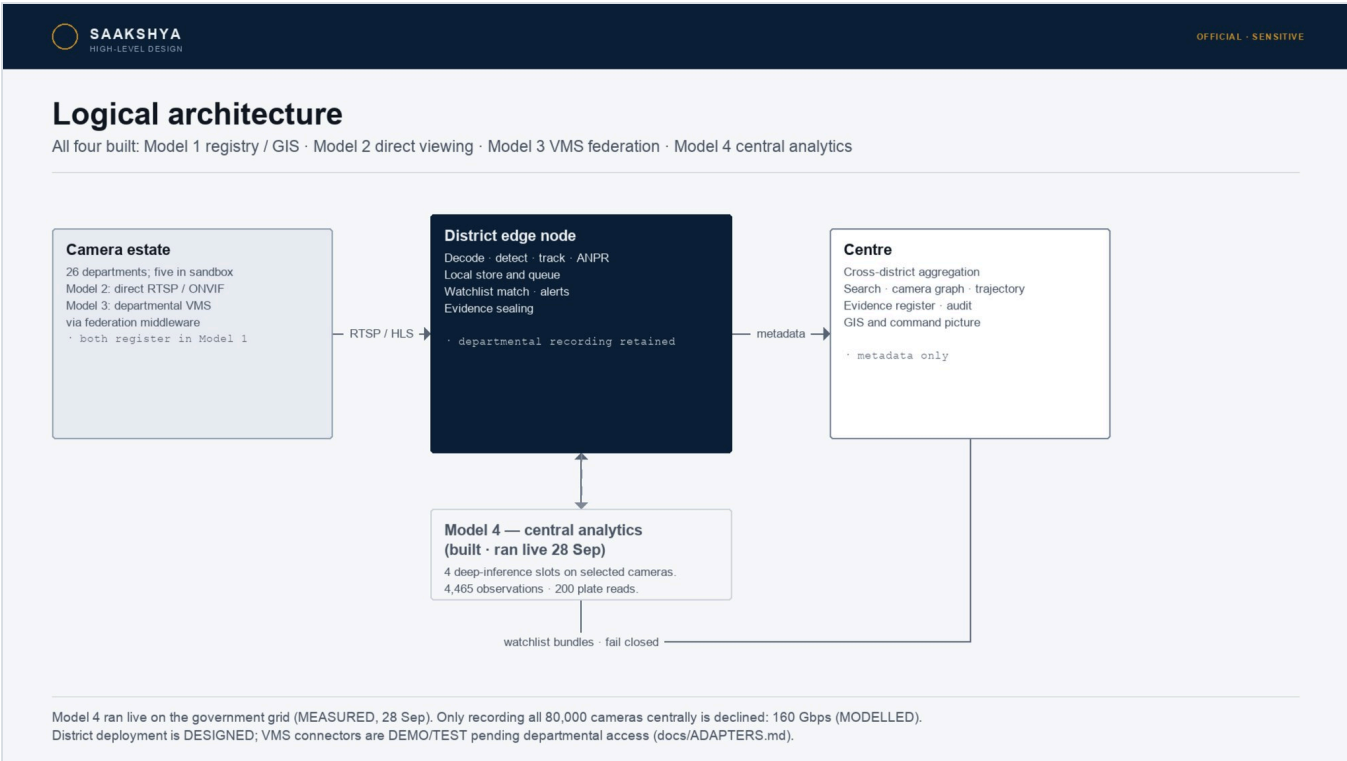
- Measure before building. We measured the government grid first: all 30 camera IDs produced RTSP frames in the census, 15 were reachable by direct WHEP and 15 needed an H.264 bridge. We also graded each camera's plate-reading capability from its own stream.
- Hybrid of all four models. Model 1 (registry and GIS) is the spine every other part reads from. Model 2 views the government grid directly through our authenticated WHEP signalling proxy, with CONTROL ROOM (30 tiles) and OPTIMIZED VIEW session policies. Model 3 federates departmental VMS systems through an adapter contract. Model 4 is central analytics on selected cameras: four deep-inference slots ranked by measured capability.
- One analytics pipeline everywhere. The steps are RT-DETRv2 detection of vehicles and people, tracking, YOLOv9 plate detection on full-resolution tiles, and Awiros-ANPR-OCR (ported to PyTorch, 6.6 ms per plate batched). A plate is published only when the frames of one track agree.
- Watchlist correlation. Each observation is matched against representative stolen, wanted, missing, blacklisted and suspect lists as it is written. Alerts are grouped into one decision per vehicle, prioritised, shown on the map, and moved through acknowledge, investigate and resolve with an audit record.
- Evidence and accountability. Every record is sealed with SHA-256 and chained to the one before; the chain is re-verified on demand. Every search needs a case and a purpose. Four authorisation gates check the token, the role, the jurisdiction and the purpose.
- Honest labels. Every number is marked MEASURED, MODELLED, DEMO or DESIGNED. Recorded footage is labelled as recorded; the synthetic route corpus is labelled as synthetic.

4. Road Map / Flow Diagram

The diagram shows the end-to-end workflow (onboard → ingest → analyse → read plates → store → match → alert → investigate → evidence) and the four-phase rollout: PoC (done), a first district cell, a first region, and statewide in department waves. Each phase opens only when its measured exit gate passes.



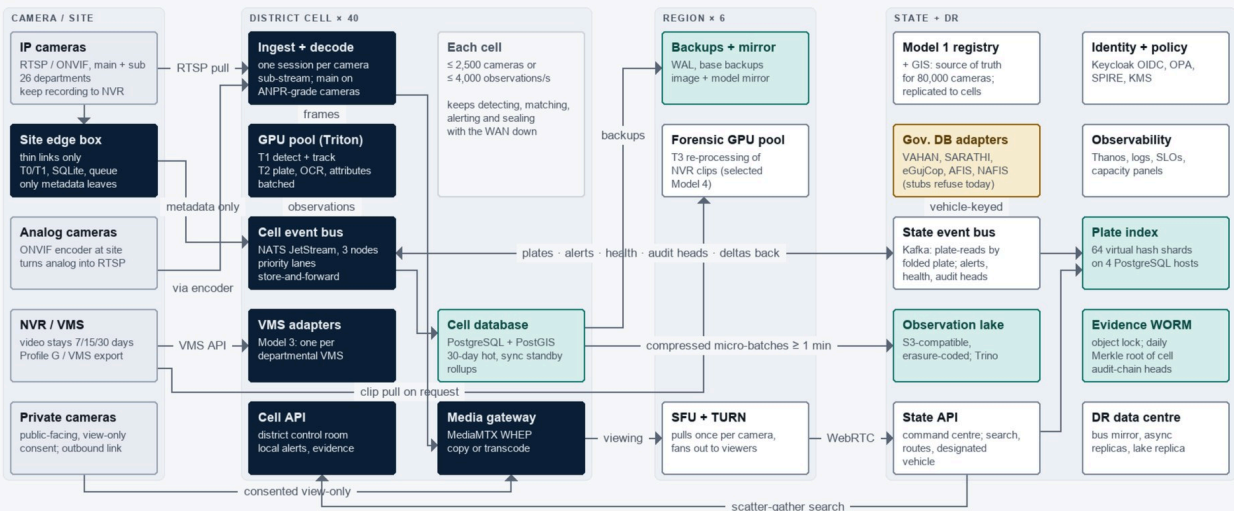
End-to-end flow and road map



Logical architecture: Models 1 + 2 + 3 + 4, all built

Statewide target architecture

Video stays where it is recorded; metadata, events, requested clips and viewed streams move



DESIGNED (docs/STATEWIDE_ARCHITECTURE.md §2-§6; HLD §3, §6, §21). Registry and watchlist deltas, DR mirroring and evidence replication are in the boxes. Only inference compute grows in proportion to cameras analysed (MODELLED, reports/capacity_model.json). Nothing here ran at 80,000 cameras.

Statewide target architecture (~80,000 cameras)

5. Tools & Technologies

- Languages: Python 3.12, JavaScript, HTML/CSS, SQL, Shell. - Backend: FastAPI and Uvicorn (REST, OpenAPI), SQLAlchemy. - Databases: SQLite for the PoC; PostgreSQL + PostGIS for the statewide design. - Video: PyAV (FFmpeg), MediaMTX (RTSP / WebRTC WHEP gateway), OpenCV. - AI: - PyTorch, ONNX Runtime and Hugging Face Transformers; - RT-DETRv2-R18 detector, YOLOv9 licence-plate detector, Awiros-ANPR-OCR (PP-OCRv5 fine-tuned on Indian plates); - fast-alpr ONNX OCR and Apple Vision as fallbacks; - Gemini 3.5 Flash on Vertex AI for the investigator copilot. - Maps: OpenStreetMap tiles and the Google Maps JavaScript API. - Security: bearer tokens, role and jurisdiction gates, purpose binding, SHA-256 hash-chained evidence and audit. - Quality: pytest (1,582 tests pass), Playwright for recorded demonstrations, a secret scanner, Git and GitHub. - Statewide design: Kubernetes district cells, NVIDIA Triton, NATS JetStream, Kafka, Ceph object storage, Trino, Keycloak and OPA.

6. Challenges & Risks

- Shared sandbox availability. During the test window only 6–13 of 30 government cameras delivered video at once, and from 28 Sep 12:57 IST the grid rejected our credentials (401). We responded with per-camera isolation, backoff, preflight gates that refuse to film a stall, and a full-feature film made on footage recorded earlier, clearly labelled. - Plate readability. Most government mounts are graded UNSUITABLE for ANPR by geometry and light; this is a camera finding, not a reader failure. Capability grades tell the state where ANPR can work. - No cross-camera government match. No plate was read on two government cameras, so there is no real multi-camera government route. Route logic is shown on a labelled synthetic corpus. - OCR errors. Of 56 plates published on a 57-second clip, 4 were wrong. The multi-frame vote and the "verify before acting" labels contain this risk. - Evidence limits. A hash proves the bytes are unchanged, not what the

image shows; older stills can show a different vehicle. The worker now seals the frame each plate was best read from. - Statewide scale. Kafka, gateway-session, TURN and GPU capacities are assumptions until the Phase 1 gates measure them. Missing department metadata (VMS, storage, retention) is itself a rollout risk. - Privacy and law. We use no face recognition on government data, blur heads in the own-feed and recorded footage, and bind every search to a case and purpose (BSA 2023, DPDP Act 2023).

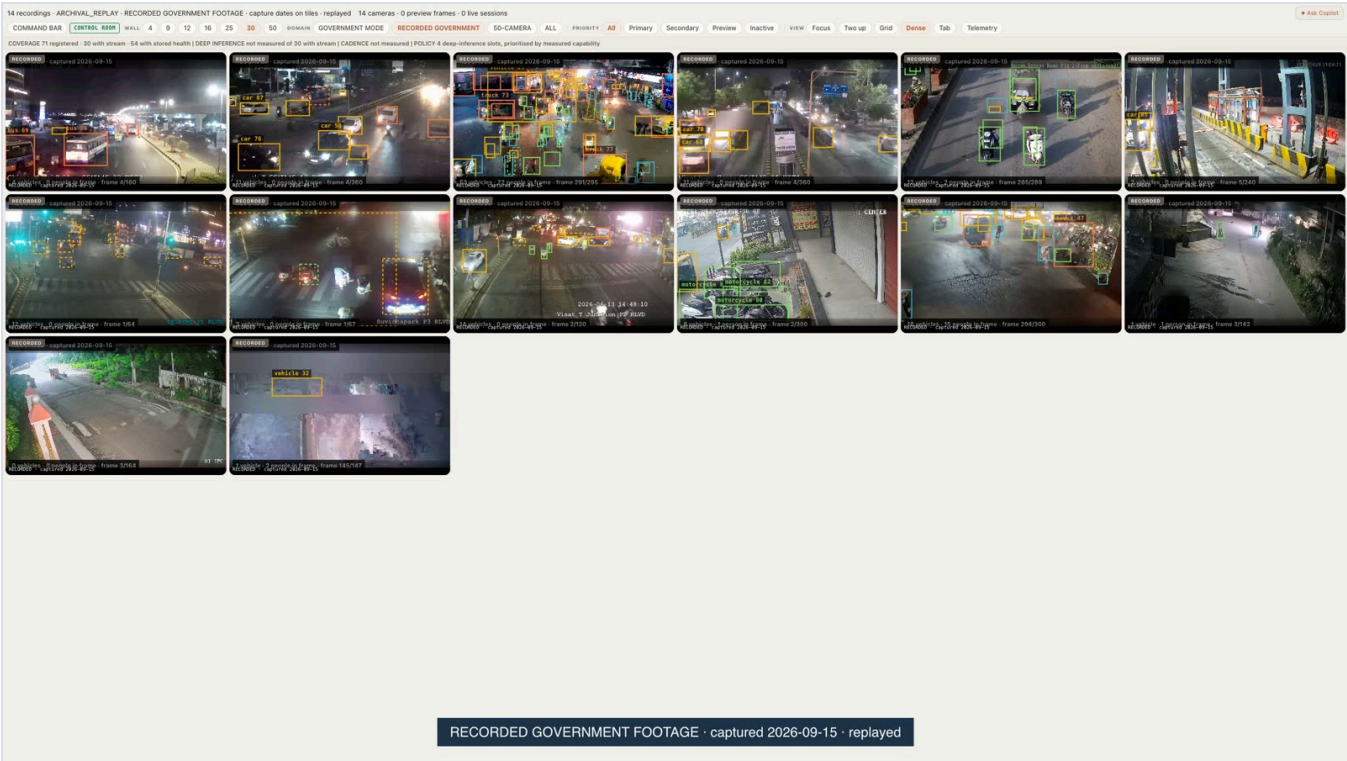
7. Possible Outcome

- One investigation screen across the state's existing cameras instead of calls to each department: a plate entered once returns every sighting with camera, time and place. - Automatic real-time alerts for stolen, wanted, missing and blacklisted vehicles, grouped so an operator makes one decision per vehicle. - Evidence that stands scrutiny: sealed, hash-chained records with a draft certificate for signature. - Affordable statewide scale: video stays local and only metadata moves; only inference compute grows with the number of cameras analysed. - A measured map of which cameras can read plates, guiding where to upgrade. - Accountability: every query attributed, purpose-bound and audited.

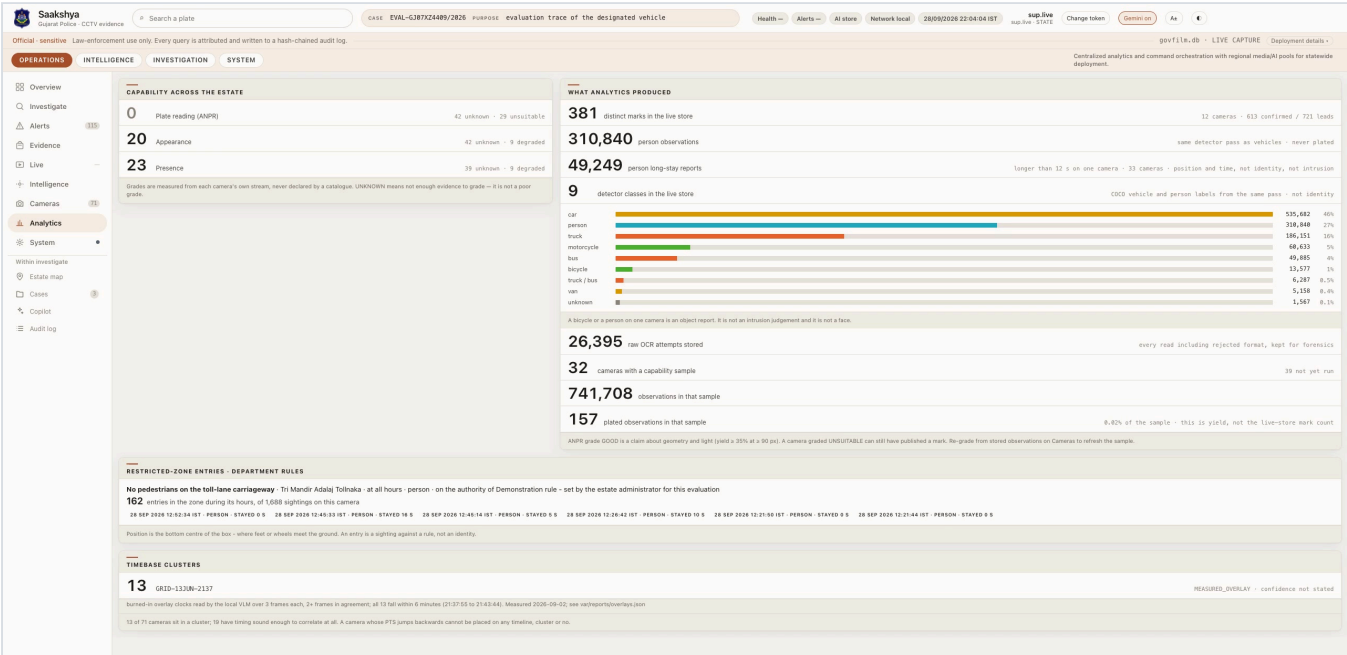
8. Work Done Till Date

- The working platform, with all four models built: - registry, GIS and onboarding (form and bulk CSV); - the live government video wall; - VMS federation adapters; - central analytics that ran live on the grid; - ANPR, watchlist alerts, search, trajectory, evidence, cases, audit and the Gemini copilot. - Measured on the government grid: - 1,155,325 observations to 24 Sep; persons on all 30 cameras, vehicles on 29; - 901 plate reads, and 200 more read live on 28 Sep; - the delivered report holds 1,101 reads with UTC and IST timestamps. - Three films: - own feed, 2:43; - government grid recorded live, 5:40; - every feature from sign-in, 15:04, on recorded government footage (49 beats, 0 failures). - Indian plate recogniser: 17 of 21 hand-read plates exact, against 5 of 21 for the previous recogniser. - Statewide design: 40 district cells, 6 regions, a state data centre and DR, with a capacity model. 80,000 synthetic registry rows were loaded and queried. - Quality: 1,582 automated tests pass; the secret scan passes on every file and the full history.

9. Solution Screenshots



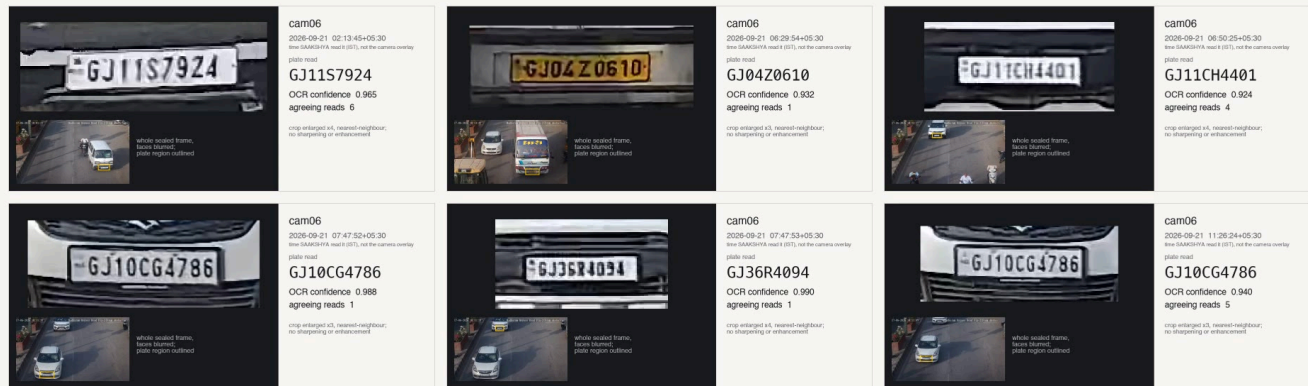
Government wall: 14 cameras with per-frame AI boxes, labelled RECORDED



Model 4 central analytics output across the estate

Government ANPR — measured observations

901 government plate reads · 178 distinct plates · 97 confirmed (≥2 agreeing reads) · 9 cameras with reads · 2026-09-02 to 2026-09-21 IST



Source: var/live.db (opened read-only) and reports/final_plate_gallery_manifest.csv. Gallery shows selected examples; statistics cover every government read in the store. Plate crops are pixel regions of the retained frames, enlarged by nearest-neighbour only. Generated 2026-09-28T11:07:31+05:30.

Government number-plate reads with confidence and agreeing reads

Saakshya
Gujarat Police - CCTV evidence

Official - sensitive Law-enforcement use only. Every query is attributed and written to a hash-chained audit log.

OPERATIONS INTELLIGENCE INVESTIGATION SYSTEM

evn/F1B-05 - LIVE CAPTURE Deployment details

Centralized analytics and command orchestration with regional media/KPI pools for statewide deployment.

OWN FEEDS: ANALYZED FRAMES BY FRAME

What the pipeline sees

Licensed street recordings from Mumbai, 2560x1440 at 30 fps. Every box is this platform's own detector and tracker on that exact frame; a registration mark is printed only once the pipeline's vote accepts it. Heads are blurred before the footage is shown.

COVERAGE 58 registered · 30 with stream · 54 with stored health · DEEP INFERENCE not measured of 30 with stream · CADENCE not measured · POLICY 4 deep-inference slots, prioritized by measured capability

OWN FEED A - STREET AND PEOPLE - OWN-MUM-AUTOSTAND Open video

OWN-MUM-AUTOSTAND RECORDED FILE: 1955x1440 30 fps. Boxes by this platform's pipeline, every frame

OWN FEED B - SIGNAL QUEUE, BEAR PLATES - OWN-MUM-QUEUE Open video

OWN-MUM-QUEUE RECORDED FILE: 1955x1440 30 fps. Boxes by this platform's pipeline, every frame

37 vehicles · 27 people in frame · frame 468/513

14 vehicles · 8 people in frame · frame 613/1716

LIVE AI PLANE

0.1

peak FPS of this API process (SME) 178.3

peak FPS of this API process (SME) 178.3

peak FPS of this API process (SME) 178.3

peak FPS of this API process (SME) 178.3

Not measured on this server: AI workers · Detector FPS · OCR FPS · Inference PSD · Inference PPS · Queue depth · GPU (show a name for why)

ESTATE RIGHT NOW

Cameras online 10

camera health status vs STREAMING

These are licensed recordings of Mumbai traffic. The browser plays each file at its native rate. The production pipeline analysed each frame beforehand. Boxes follow the video's frame clock. Plates appear when the pipeline's agreement vote holds. This is analysis replay, not a claim of live inference speed.

Not measured on this server: Active WHEP · Preview · AI streams · AI latency PSD · AI latency PPS · Watchlist alert latency (show a name for why)

Own feed: vehicles boxed, plates drawn only when the vote holds

Saakshya Search a plate CASE: FIR-000/2026 PURPOSE: reviewing government camera observations Online 26 · degraded 35 (store) Alerts 141 AI DEGRADED · OCR DEGRADED Network local 28/09/2026 21:55:25 IST sup.Riv govt11a-05 LIVE CAPTURE Deployment details

REPRESENTATIVE ALERT – ACKNOWLEDGE

OPERATIONS INTELLIGENCE INVESTIGATION SYSTEM

Overview Investigate Alerts 133 Evidence Live Intelligence Cameras Analytics System

Written investigate Estate map Cases Copilot Audit log

ALERTS
142 alerts 13 vehicles — grouped so each vehicle is one decision. Reads within 10 minutes of each other are one pass. J/K move, A acknowledges, X resolves.

GJ01R59114 EXACT
designated vehicle of interest (evaluation)
READ **GJ01R59114**
LISTED **GJ01R59114** EXACT
plate agreed across 3 frames
10 reads · 1 camera · Timbavadi Gate, Junagadh
first 21 Sep 00:50:38 IST · 7 d ago · last 28 Sep 12:50:28 IST · 9 h ago
Recommended: **VERIFY and notify the district control room**
INVESTIGATE Trace report View video
Acknowledge at 18 Resolve OPEN 14

GJ11B0928 EXACT
designated vehicle of interest (evaluation)
READ **GJ11B0928**
LISTED **GJ11B0928** EXACT
plate agreed across 11 frames
26 reads · 1 camera · Timbavadi Gate, Junagadh
first 20 Sep 09:52:00 IST · 8 d ago · last 28 Sep 12:47:12 IST · 9 h ago
Registration mark read on a single frame — an exact match but not corroborated across frames, verify before acting
Recommended: **VERIFY and notify the district control room**
INVESTIGATE Trace report View video
Acknowledge at 24 Resolve ACKNOWLEDGED 1 · OPEN 24

GJ1159258 EXACT
designated vehicle of interest (evaluation)
READ **GJ1159258** LISTED **GJ1159258**
plate agreed across 7 frames
20 reads · 1 camera · Timbavadi Gate, Junagadh
first 21 Sep 00:42:30 IST · 7 d ago · last 28 Sep 12:42:03 IST · 9 h ago
Registration mark read on a single frame — an exact match but not corroborated across frames, verify before acting
Recommended: **VERIFY and notify the district control room**
INVESTIGATE Trace report View video
Acknowledge at 22 Resolve OPEN 22

GJ18Z8826 EXACT
designated vehicle of interest (evaluation)
READ **GJ18Z8826** LISTED **GJ18Z8826**
plate agreed across 2 frames
2 reads · 1 camera · Timbavadi Gate, Junagadh
first 21 Sep 12:54:14 IST · 7 d ago · last 28 Sep 12:54:45 IST · 9 h ago
Recommended: **VERIFY and notify the district control room**
INVESTIGATE Trace report View video
Acknowledge at 2 Resolve OPEN 5

GJ11B2886 EXACT
designated vehicle of interest (evaluation)
READ **GJ11B2886** EXACT
plate agreed across 6 frames
19 reads · 1 camera · Timbavadi Gate, Junagadh
first 21 Sep 00:49:28 IST · 7 d ago · last 28 Sep 12:27:04 IST · 9 h ago
Registration mark read on a single frame — an exact match but not corroborated across frames, verify before acting
Recommended: **VERIFY and notify the district control room**
INVESTIGATE Trace report View video
Acknowledge at 18 Resolve OPEN 18

GJ1157924 EXACT
designated vehicle of interest (evaluation)
READ **GJ1157924** LISTED **GJ1157924**
plate agreed across 13 frames
56 reads · 1 camera · Timbavadi Gate, Junagadh
first 20 Sep 20:57:55 IST · 7 d ago · last 28 Sep 12:20:30 IST · 9 h ago
Recommended: **VERIFY and notify the district control room**
INVESTIGATE Trace report View video null
Acknowledge at 26 Resolve ACKNOWLEDGED 26

GJ18X6705 EXACT
designated vehicle of interest (evaluation)
READ **GJ18X6705** LISTED **GJ18X6705**
plate agreed across 6 frames
71 reads · 1 camera · Timbavadi Gate, Junagadh
first 20 Sep 09:45:45 IST · 8 d ago · last 28 Sep 12:19:08 IST · 9 h ago
Registration mark read on a single frame — an exact match but not corroborated across frames, verify before acting
Recommended: **VERIFY and notify the district control room**
INVESTIGATE Trace report View video
Acknowledge at 16 Resolve OPEN 16

GJ18Z1782 EXACT
designated vehicle of interest (evaluation)
READ **GJ18Z1782** EXACT
plate agreed across 4 frames
10 reads · 1 camera · Timbavadi Gate, Junagadh
first 21 Sep 00:52:50 IST · 7 d ago · last 21 Sep 12:52:54 IST · 7 d ago
Registration mark read on a single frame — an exact match but not corroborated across frames, verify before acting
Recommended: **VERIFY and notify the district control room**
INVESTIGATE Trace report View video
Acknowledge at 10 Resolve OPEN 10

GJ11B1008 EXACT
designated vehicle of interest (evaluation)
READ **GJ11B1008** EXACT
plate agreed across 2 frames
2 reads · 1 camera · Timbavadi Gate, Junagadh
first 21 Sep 03:25:27 IST · 7 d ago · last 21 Sep 06:50:20 IST · 7 d ago
Registration mark read on a single frame — an exact match but not corroborated across frames, verify before acting
Recommended: **VERIFY and notify the district control room**
INVESTIGATE Trace report View video
Acknowledge at 2 Resolve OPEN 2

GJ25W7686 EXACT
designated vehicle of interest (evaluation)
READ **GJ25W7686** LISTED **GJ25W7686**
plate agreed across 5 frames
3 reads · 1 camera · Timbavadi Gate, Junagadh
first 21 Sep 02:57:52 IST · 7 d ago · last 21 Sep 04:59:34 IST · 7 d ago
Recommended: **VERIFY and notify the district control room**
INVESTIGATE Trace report View video
Acknowledge at 3 Resolve OPEN 3

GJ11B09063 EXACT
designated vehicle of interest (evaluation)
READ **GJ11B09063** EXACT
plate agreed across 2 frames
1 read · 1 camera · Timbavadi Gate, Junagadh
first 20 Sep 08:46:25 IST · 8 d ago · last 20 Sep 08:46:25 IST · 8 d ago
Registration mark read on a single frame — an exact match but not corroborated across frames, verify before acting
Recommended: **VERIFY and notify the district control room**
INVESTIGATE Trace report View video Acknowledge
Acknowledge at 1 Resolve OPEN 1

GJ38BH5815 EXACT
designated vehicle of interest (evaluation)
READ **GJ38BH5815** EXACT
plate agreed across 2 frames
1 read · 1 camera · Dethal Char Rasta, Patan
first 02 Sep 06:56:22 IST · 26 d ago · last 02 Sep 06:56:22 IST · 26 d ago
Registration mark read on a single frame — an exact match but not corroborated across frames, verify before acting
Recommended: **VERIFY and notify the district control room**
INVESTIGATE Trace report View video Acknowledge
Acknowledge at 1 Resolve OPEN 1

GJ11Y6198 EXACT
designated vehicle of interest (evaluation)
READ **GJ11Y6198** EXACT
plate agreed across 2 frames
2 reads · 1 camera · Timbavadi Gate, Junagadh
first 19 Sep 05:17:58 IST · 7 d ago · last 19 Sep 11:56:33 IST · 7 d ago

57 cameras unbracketed
0 graded good for plate reading
0 marks read in the latest hour
141 alerts open, unacknowledged

A representative demonstration incident. The transition is recorded in the audit log.

Alert queue grouped per vehicle

Saakshya Search a plate CASE: FIR-000/2026 PURPOSE: demonstrating review of government observations Online 26 · degraded 35 (store) Alerts 141 AI DEGRADED · OCR DEGRADED Network local 28/09/2026 21:51:38 IST sup.Riv govt11a-05 LIVE CAPTURE Deployment details

Official: sensitive Law-enforcement use only. Every query is attributed and written to a hash-chained audit log.

OPERATIONS INTELLIGENCE INVESTIGATION SYSTEM

Overview Investigate Alerts 133 Evidence Live Intelligence Cameras Analytics System

Written investigate Estate map Cases Copilot Audit log

LIVE 14 recordings ARCHIVAL · REPLAY · RECORDED GOVERNMENT FOOTAGE · capture dates on tiles · replayed 14
cameras · 0 preview frames · 0 live sessions

ANALYTICS OFF VEHICLES PEOPLE VEHICLES · PEOPLE ANPR FULL INCIDENT OVERLAY REFRESH FAST BALANCED DEEP FILTER ALL VEHICLES PEOPLE PEOPLE VEHICLES ANPR SHOW TRACKS COMPARE
PRESET CONTROL ROOM 3D-CAMERA OVERVIEW 3D-CAMERA WALL INTELLIGENCE DEMO TRAFFIC INTELLIGENCE PERSON SEARCH WATCHLIST INCIDENT DESIGNATED VEHICLE

People: 0 · Vehicles: 0 · Tracked: 0

All Ahmedabad Gandhinagar Junagadh Rajkot Navsari Gujarat (town not confirmed) Patan Unlocated

8 vehicles · 1 person in frame · frame 54203

ARCHIVAL · REPLAY · Plates read on GOVREC-cam06 · select a plate for its route

RECORDED GOVERNMENT FOOTAGE · captured 2026-09-15 · replayed · intelligence panel

District Junagadh
Department Home (Police)
Source ARCHIVAL · REPLAY
Recorded from cam06
Capture window 2026-09-15T06:30:45+05:30
Capture window 2026-09-15T06:31:01+05:30
Stream h264 · 1920x1080
Measured FPS not measured
AI state not measured
Basis derived from name

RECENT OBSERVATIONS
People: 0 · Vehicles: 0 · Tracked: 0

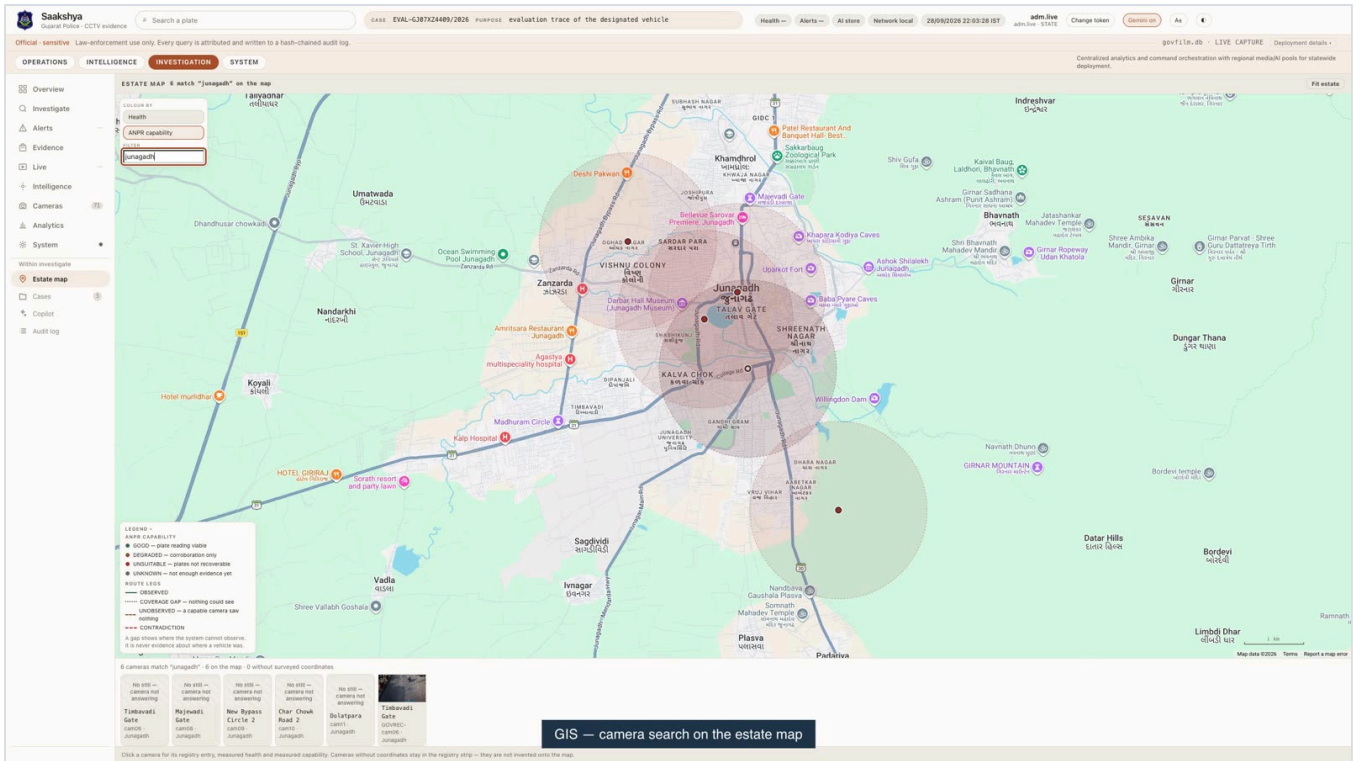
Measured capability
Position derived from the camera's name. Not a narrowed problem — good enough to reason about a corridor, never evidence of where a vehicle was.

EVIDENCED NEIGHBOURS
No transition with a sighting behind it is listed.

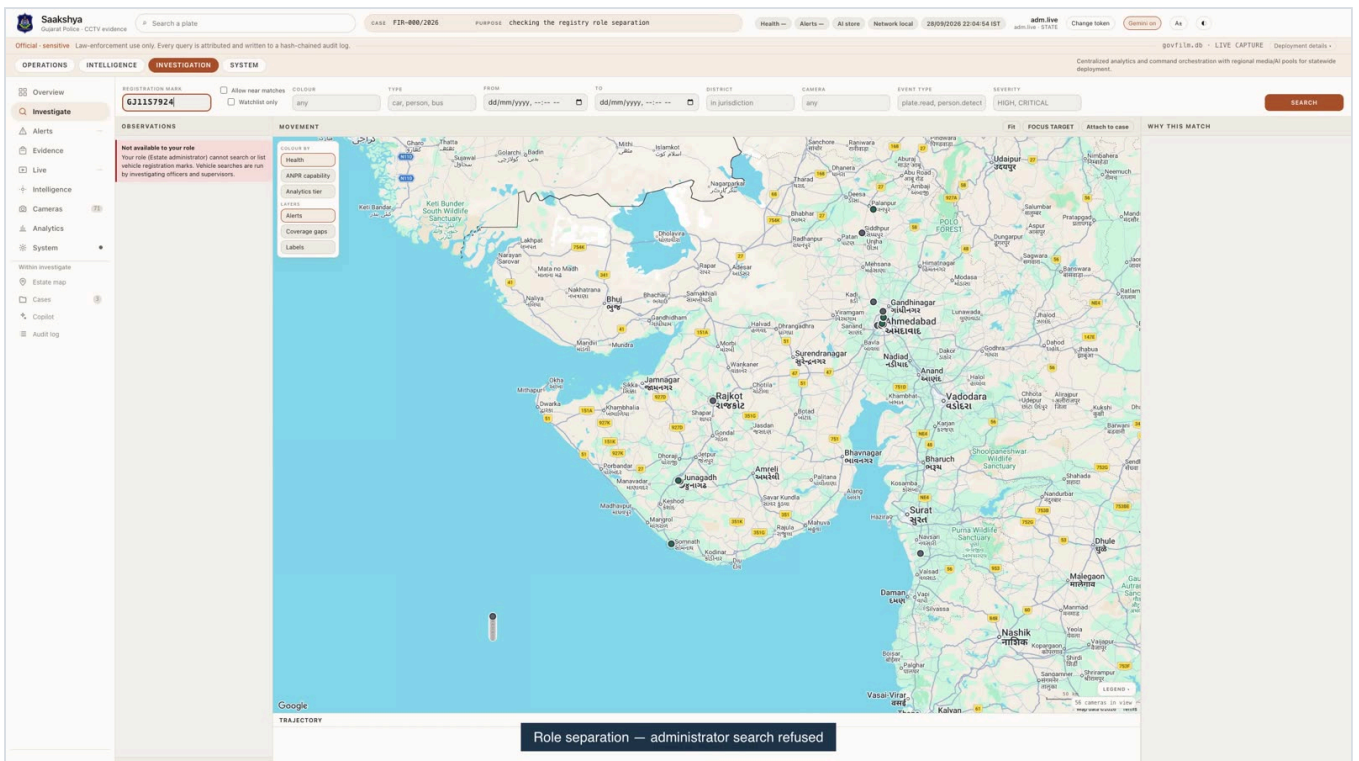
Focused camera with provenance panel

Evidence chain verification

Gemini copilot, grounded in stored results



Estate map with camera search



Role separation: administrator search refused

10. Project Report

This document. Films, deck, HLD, plate reports and evidence records are in the repository's submission/ folder and the v1.0 release.